



University
Mohammed VI
Polytechnic



Deliverable #: Title of Lab/Project here (e.g.: Conceptual Design)

Data Management Course
UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	
Member 1	Malak Nhami
Member 2	Marwa Mazouz
Member 3	Mohamed Moussaid
Member 4	Sara Lamnaïy
Repository Link	https://github.com/malak-3108/project

1 Introduction

This deliverable is the second step of the RLMS project, following up on the conceptual design (ER diagram) built in Lab 1. The main challenge here is turning that diagram into an actual database structure — that is, deciding what tables are needed, what columns go in each one, and how the tables connect to each other. This report explains the tables we created, the keys used to identify and link records, and the reasoning behind the choices we made when converting more complex parts of the diagram, such as ISA hierarchies, weak entities, and ternary relationships.

2 Requirements

- 1 - Convert the lab1 ER diagram into relational tables (entities + relationships).
- 2 - For each table: name, attributes, primary key, foreign keys, and any candidate keys.
- 3 - Handle special ER cases: ISA hierarchies, weak entities, recursive relationships, ternary relationships, relationships with attributes, and aggregation.
- 4 - Note any constraints that keys alone can't capture.

3 Methodology

ISA Hierarchies: Each subclass is its own table whose primary key is also a foreign key to the parent's key, keeping subtype-specific attributes separate without duplicating shared ones.

Weak Entities (Maintenance and Calibration): Both are owned by EquipmentUnit. Their primary keys combine the owner's key with a partial key: (maintainedSerialno, startTS) and (calibSerialno, calibdate). Each references equipmentUnit(serialno) with ON DELETE CASCADE to preserve the identifying dependency.

Ternary Relationship (Supplies): Connects Supplier, Consumable, and Laboratory, since unitPrice belongs to the triple. Mapped as one table with a composite key of all three foreign keys.

Aggregation (Monitors): Monitors relates Technical to the aggregated Stocks relationship, not to Consumable/Laboratory directly. Implemented by adding monitored-byTechnicalId as a foreign key inside stocks, which already holds the aggregated pair's key.

4 Implementation & Results

ISA Hierarchies(one exemple):

```

1 create table student (
2     student_id int,
3     program varchar(20),
4     primary key(student_id),
5     foreign key (student_id) references person(pid)

```

6);

Ternary Relationship (Supplies):

```

1 create table supplies (
2     suppid int,
3     consid int,
4     labid int,
5     unitprice decimal,
6     primary key (suppid, consid, labid),
7     foreign key (consid) references consumable(consid),
8     foreign key (suppid) references supplier(suppid),
9     foreign key (labid) references laboratory(labid)
10 );

```

Aggregation (Monitors):

```

1 create table stocks (
2     monitoredby_technical_id int not null,
3     consid int,
4     labid int,
5     ...
6     primary key (consid, labid),
7     foreign key (monitoredby_technical_id) references technical(
8         technical_id) on delete no action
9 );

```

5 Discussion

The main challenge was mapping the aggregation and weak entities while preserving integrity constraints, especially placing primary keys correctly for the aggregation and ternary relationship. We learned that translating an ER diagram into a relational schema requires careful, deliberate decisions, as small choices can affect how well the schema preserves the original semantics.

6 Conclusion

The deliverable successfully translates the RLMS ER diagram into a working relational schema, implemented as SQL CREATE TABLE statements that preserve the entities, relationships, and constraints defined in the conceptual design.